

Problem 1: 95% Confidence Interval for True Error

Given

$n = 45$ test instances, sample error $\text{error}_S(h_1) = 6.67\% = 3/45 = 0.0667$ (3 misclassified; h_1 confusion matrix: 1 FP + 2 FN). The 'training data' in the question refers to the 45-instance held-out test set.

Confidence level 95% $\rightarrow z_N = 1.96$ (Table 3).

Formula

$$\text{error}_D(h) = \text{error}_S(h) \pm z_N \cdot \sqrt{(\text{error}_S(h)(1 - \text{error}_S(h)) / n)}$$

(Normal approximation valid: $n \cdot e \cdot (1-e) = 2.80 \approx 3$ is borderline, but the assignment expects this method.)

Calculation

$$\text{Standard error} = \sqrt{(0.0667 \times 0.9333 / 45)} = \sqrt{0.001383} = 0.0372$$

$$\text{Margin} = 1.96 \times 0.0372 = 0.0729$$

$$\text{Interval} = 0.0667 \pm 0.0729 = [-0.0062, 0.1395]$$

Lower bound is negative, so it is clipped to 0 (error cannot be below 0).

Answer: 95% CI for the true error of $h_1 \approx [0, 0.1395]$ i.e. 0% to 14.0%.